

Common-Mode Voltage Reduction in Two-Level Inverters by Introducing Adaptive Carrier-Phase-Shift Method

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Introduction & Motivation

Two-Level Voltage Source Inverters dominate industrial motor drives due to simplicity, low cost, and reliability.

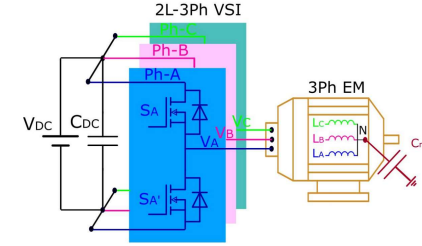


Fig.1: The system structure of a 2L-3Ph motor drive system.

Problem: PWM outputs → Common-Mode Voltage → leads to:

- Electromagnetic Interference (EMI).
- Bearing currents → insulation damage → reduced lifespan.

Existing Mitigation Approaches:

- New inverter topologies / CM filters → ↑ cost & complexity.
- Constant carrier-phase-shift PWM → partial CMV reduction only.

Key Idea:

- Dynamically adjusts carrier phase shifts every switching interval.
- Minimizes CMV harmonics at the instantaneous level.
- No hardware changes → scalable & low-cost.

Analytical Investigation of CMV

- CMV in 2L-VSI:

$$V_{CM} = \frac{V_A + V_B + V_C}{3}$$

- When conventional single-carrier SPWM is implemented, the common-mode voltage (V_{CM}) can be calculated as:

$$V_{CM} = \frac{V_{DC}}{2} + \frac{2V_{DC}}{\pi} \sum_{m=1}^{m=\infty} \left[\frac{J_0\left(m\frac{\pi}{2}\right) \sin\left(m\frac{\pi}{2}\right)}{\cos\left(m\omega_c t\right)} \right] + \frac{2V_{DC}}{\pi} \sum_{m=1}^{m=\infty} \sum_{n=-\infty}^{n=\infty} \left[\frac{\frac{1}{m} J_n\left(m\frac{\pi}{2}\right) \sin\left((m+n)\frac{\pi}{2}\right)}{\cos\left(m\omega_c t + n\omega_r t\right)} \right]$$

$n \equiv 0 \pmod{3}$
($m+n$) is not even

Harmonic decomposition → Zero-sequence components create CM currents.

Table. 1 : Frequency Components and Their Sequences Around First Carrier Harmonic.

Frequency	Phase-A	Phase-B	Phase-C	Sequence
f_0	0°	120°	-120°	Positive
$f_s - 6f_0$	0°	0°	0°	Zero
$f_s - 4f_0$	0°	-120°	120°	Negative
$f_s - 2f_0$	0°	120°	-120°	Positive
f_s	0°	0°	0°	Zero
$f_s + 2f_0$	0°	-120°	120°	Negative
$f_s + 4f_0$	0°	120°	-120°	Positive
$f_s + 6f_0$	0°	0°	0°	Zero

Conventional Interleaving

- Shifts carriers by 0° , 120° , 240° → redistributes harmonics.

Table. 2 : Frequency Components and Their Phases Around First Carrier Harmonic for Multi-Carrier Signals.

Frequency	Phase-A	Phase-B	Phase-C	Sequence
f_0	0°	120°	$-120^\circ + \phi_{CC}$	Positive
$f_s - 6f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$0^\circ + \phi_{CB} = -120^\circ$	$0^\circ + \phi_{CC} = 120^\circ$	Negative
$f_s - 4f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$-120^\circ + \phi_{CB} = 120^\circ$	$-120^\circ + \phi_{CC} = -120^\circ$	Positive
$f_s - 2f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$120^\circ + \phi_{CB} = 0^\circ$	$-120^\circ + \phi_{CC} = 0^\circ$	Zero
f_s	$0^\circ + \phi_{CA} = 0^\circ$	$0^\circ + \phi_{CB} = -120^\circ$	$0^\circ + \phi_{CC} = 120^\circ$	Negative
$f_s + 2f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$-120^\circ + \phi_{CB} = 120^\circ$	$120^\circ + \phi_{CC} = -120^\circ$	Positive
$f_s + 4f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$120^\circ + \phi_{CB} = 0^\circ$	$0^\circ + \phi_{CC} = 0^\circ$	Zero
$f_s + 6f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$0^\circ + \phi_{CB} = -120^\circ$	$0^\circ + \phi_{CC} = 120^\circ$	Negative

$$\phi_{CA} = 0, \phi_{CB} = -120^\circ, \text{ and } \phi_{CC} = +120^\circ.$$

Proposed Method

Represent 1st carrier harmonics of each phase in phasor domain.

$$V_A(t)^{\omega_c} = \frac{2V_{DC}}{\pi} \sin(\pi D_A) \cos(\omega_c t + \phi_{CA}),$$

$$V_B(t)^{\omega_c} = \frac{2V_{DC}}{\pi} \sin(\pi D_B) \cos(\omega_c t + \phi_{CB}),$$

$$V_C(t)^{\omega_c} = \frac{2V_{DC}}{\pi} \sin(\pi D_C) \cos(\omega_c t + \phi_{CC}),$$

Adjust phase shifts to either:

- Form a triangle → cancel CMV completely.
- Align small vectors opposite to the largest vector (triangle inequality not met).

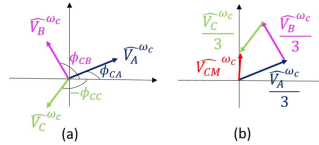


Fig.2: Phasor domain analysis of the CMV in the first switching harmonic. a) The phase voltages. b) The CMV.

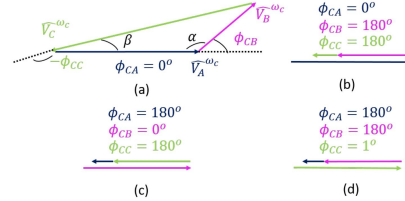


Fig.3: Determination of the carrier phase shifts in phasor domain analysis. a) Triangle formation. b) Phase-A voltage vector is bigger than other vectors. c) Phase-B voltage vector is bigger than other vectors. d) Phase-C voltage vector is bigger than other vectors.

Algorithm

Algorithm 1 Determination of the Carrier Phase Shifts

- 1: if $|\hat{V}_A^{\omega_c} - \hat{V}_B^{\omega_c}| < \hat{V}_C^{\omega_c} < |\hat{V}_A^{\omega_c} + \hat{V}_B^{\omega_c}|$ then
- 2: Take $\phi_{CA} = 0^\circ$;
- 3: Apply cosine theorem to find ϕ_{CB} :

$$(\hat{V}_A^{\omega_c})^2 + (\hat{V}_B^{\omega_c})^2 - 2\hat{V}_A^{\omega_c}\hat{V}_B^{\omega_c}\cos(\alpha) = (\hat{V}_C^{\omega_c})^2$$

$$\cos[\alpha] = \frac{(\hat{V}_A^{\omega_c})^2 + (\hat{V}_B^{\omega_c})^2 - (\hat{V}_C^{\omega_c})^2}{2\hat{V}_A^{\omega_c}\hat{V}_B^{\omega_c}}$$

$$\alpha = \arccos\left[\frac{(\hat{V}_A^{\omega_c})^2 + (\hat{V}_B^{\omega_c})^2 - (\hat{V}_C^{\omega_c})^2}{2\hat{V}_A^{\omega_c}\hat{V}_B^{\omega_c}}\right]$$

$$\phi_{CB} = 180^\circ - \alpha$$

- 4: Apply sine theorem to find ϕ_{CC} :

$$\frac{\sin[\beta]}{\hat{V}_B^{\omega_c}} = \frac{\sin[\alpha]}{\hat{V}_C^{\omega_c}}$$

$$\sin[\beta] = \frac{\hat{V}_B^{\omega_c} \sin[\alpha]}{\hat{V}_C^{\omega_c}}$$

$$\beta = \arcsin\left[\frac{\hat{V}_B^{\omega_c} \sin[\alpha]}{\hat{V}_C^{\omega_c}}\right]$$

$$\phi_{CC} = \beta - 180^\circ$$

- 5: else
- 6: if $|\hat{V}_A^{\omega_c} > \hat{V}_B^{\omega_c}|$ and $|\hat{V}_A^{\omega_c} > \hat{V}_C^{\omega_c}|$ then
- 7: $\phi_{CA} = 0^\circ$, $\phi_{CB} = 180^\circ$ and $\phi_{CC} = 180^\circ$
- 8: else if $|\hat{V}_B^{\omega_c} > \hat{V}_A^{\omega_c}|$ and $|\hat{V}_B^{\omega_c} > \hat{V}_C^{\omega_c}|$ then
- 9: $\phi_{CA} = 180^\circ$, $\phi_{CB} = 0^\circ$ and $\phi_{CC} = 180^\circ$
- 10: else if $|\hat{V}_C^{\omega_c} > \hat{V}_A^{\omega_c}|$ and $|\hat{V}_C^{\omega_c} > \hat{V}_B^{\omega_c}|$ then
- 11: $\phi_{CA} = 180^\circ$, $\phi_{CB} = 180^\circ$ and $\phi_{CC} = 0^\circ$
- 12: end if
- 13: end if

Discussion

- **Bearing current mechanisms:** capacitive, electric discharge machining (EDM), and circulating bearing currents.
- **Capacitive bearing currents:** strongly influenced by high-frequency CMV components; higher dv/dt → larger currents.
- **EDM currents:** triggered when CMV exceeds lubricant breakdown voltage → electrical discharge across bearings.
- **Switching pattern impact:** less critical for capacitive currents but important for EDM current suppression.

Simulation Results

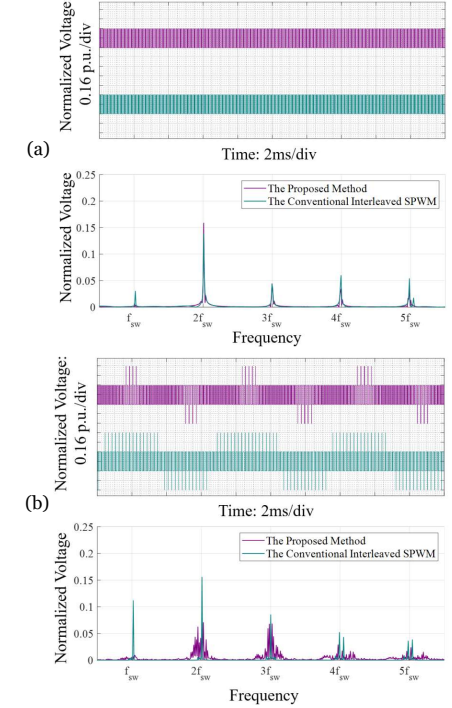


Fig.4: The CMVs and their FFTs for two modulation indices. a) ma = 0.4. b) ma = 0.8.

Experimental Results

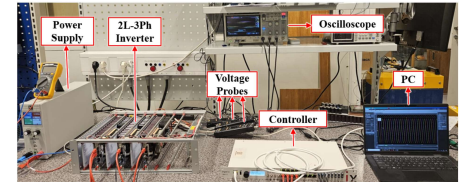


Fig.5: Experimental setup of 2-level 3-phase inverter system.

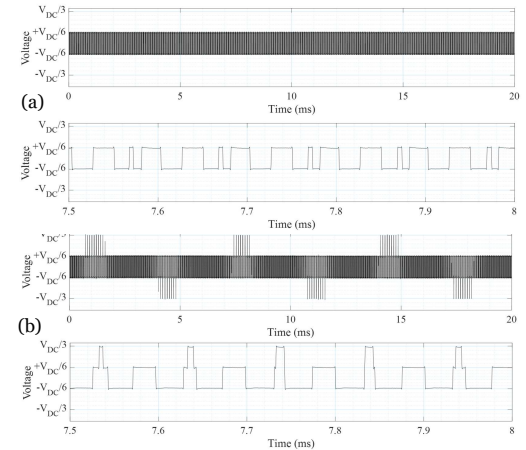


Fig.6: CMVs for two modulation indices. a) ma = 0.4. b) ma = 0.8. (at $f_0 = 50$ and $f_s = 10\text{kHz}$).

Conclusion

Adaptive Carrier-Phase-Shift SPWM:

- Reduces instantaneous CMV more effectively than constant carrier phase-shift PWMs.
- Maintains simplicity of 2L-VSIs.
- No additional hardware required.

Future Work:

- Extend to multi-phase & multi-level inverters.